
THE VELOX RESEARCHER'S HANDBOOK

Reconstitution, Storage & Documentation

A practical reference for handling research peptides in the lab

VELOX PEPTIDES

UK & EU research peptides · veloxpeps.com

What Research Peptides Are

A **peptide** is a short chain of amino acids — the same building blocks that make up proteins, joined together in a defined sequence. Where proteins are large and complex, peptides are smaller and more precisely defined, which makes them useful reference materials in laboratory work.

Velox research peptides are supplied as a **lyophilised (freeze-dried) powder** sealed in a glass vial. Lyophilisation removes water from the material, leaving a stable solid that keeps far longer than a liquid would. Before the material can be used in solution-based work, it is **reconstituted** — dissolved back into a liquid — a process covered in the next section.

Important — intended use

All Velox peptides are sold strictly for **in vitro research use only**. They are not drugs, supplements or medicines, and are not for human or veterinary consumption, injection, dosing, diagnosis or treatment. This handbook describes laboratory technique, quality and documentation only.

At a glance

Property	Velox research peptides
Physical form	Lyophilised powder in a sealed glass vial
Purity	≥99% by HPLC
Testing	Third-party tested by Janoshik Analytical
Documentation	Certificate of Analysis (CoA) supplied per batch
Intended use	In vitro laboratory research only

Reconstitution Basics

Reconstitution is the process of dissolving the lyophilised powder back into a liquid so it can be handled in solution. The most common diluent for this purpose is **bacteriostatic water** — sterile water containing a small amount of benzyl alcohol (typically 0.9%), which suppresses bacterial growth and allows a reconstituted vial to be used over a longer working period than plain sterile water.

The Concentration Formula

Concentration depends only on the mass of peptide in the vial and the volume of liquid you add:

$$\text{Concentration (mg/mL)} = \text{mass in vial (mg)} \div \text{water added (mL)}$$

Worked example

Suppose a vial contains **10 mg** of peptide and you add **2 mL** of bacteriostatic water:

$$10 \text{ mg} \div 2 \text{ mL} = 5 \text{ mg/mL}$$

So every 1 mL of the resulting solution contains 5 mg of peptide; every 0.1 mL contains 0.5 mg.

Quick-reference: concentration by vial size and water volume (mg/mL)

Vial size	+1 mL	+2 mL	+3 mL	+5 mL
5 mg	5.0	2.5	1.67	1.0
10 mg	10.0	5.0	3.33	2.0
15 mg	15.0	7.5	5.0	3.0
20 mg	20.0	10.0	6.67	4.0
30 mg	30.0	15.0	10.0	6.0

Skip the arithmetic

Use the free Velox **reconstitution calculator** to work out concentration for any vial size and water volume instantly: veloxpeps.com/tools/reconstitution-calculator/

Storage & Handling

Correct storage preserves the integrity of a research peptide. The two states — dry lyophilised powder and reconstituted solution — have different requirements.

Storage conditions by state

State	Temperature	Notes
Lyophilised powder	-20°C	Store dry and dark; keep the vial sealed. Stable for extended periods. Short transit at ambient temperature is normally well tolerated.
Reconstituted solution	2–8°C (refrigerated)	Keep refrigerated and protected from light. Avoid repeated freeze–thaw cycles. Use within weeks rather than months.

Sterile technique

- Wipe the vial septum with an alcohol swab before and after drawing liquid.
- Introduce the diluent slowly down the inside wall of the vial; let the powder dissolve gently rather than shaking vigorously.
- Use clean, single-use consumables and avoid touching surfaces that contact the solution.
- Label every reconstituted vial with the compound, concentration and date prepared.

Signs of degradation

Inspect solutions before use. Discard any vial showing the following, as they can indicate that the material has degraded or become contaminated:

- Cloudiness, haze or visible particulates in a solution that should be clear.
- Unexpected colour change.
- Precipitate that does not redissolve.
- Powder that has visibly clumped, melted or changed colour before reconstitution.

How to Read a Certificate of Analysis

Every Velox batch ships with a **Certificate of Analysis (CoA)** — an independent test report describing exactly what is in the vial. Knowing how to read it lets you verify quality for yourself rather than taking it on trust.

What the CoA reports

Field	What it tells you
HPLC purity %	The proportion of the sample that is the target peptide, measured by High-Performance Liquid Chromatography. Velox targets $\geq 99\%$. A higher figure means fewer impurities.
Mass spectrometry	Confirms the molecular weight of the peptide matches its expected value — evidence that the correct compound was synthesised.
Appearance	A physical description of the material (for example, white lyophilised powder), giving a baseline to compare your vial against.

What "PASS" means

A result marked **PASS** indicates the batch met the predefined specification for that test — for example, purity at or above the threshold and molecular weight within tolerance. A PASS across all tests means the batch is consistent with the stated quality standard.

Why third-party testing matters

Velox CoAs are produced by **Janoshik Analytical**, an independent laboratory with no commercial stake in the result. Independent verification is more credible than a supplier testing its own product, because the lab running the assay has no incentive to flatter the outcome. Each CoA is tied to a specific batch, so you can match the document to the vial in front of you.

Mini-Glossary

Term	Definition
Lyophilised	Freeze-dried. Water is removed under vacuum to leave a stable dry powder.
Reconstitution	Dissolving a lyophilised powder back into a liquid so it can be handled in solution.
Bacteriostatic water	Sterile water with a small amount of benzyl alcohol added to inhibit bacterial growth.
HPLC	High-Performance Liquid Chromatography — an analytical method used to measure purity.
Mass spectrometry	A technique that measures molecular weight to confirm a compound's identity.
Molecular weight	The mass of one molecule of the peptide; a fingerprint used to confirm identity.
CoA	Certificate of Analysis — a per-batch report documenting test results.
Purity	The proportion of a sample that is the intended compound, expressed as a percentage.
In vitro	Latin for "in glass" — laboratory work performed outside a living organism.
Vial	The small sealed glass container in which the lyophilised peptide is supplied.

VELOX PEPTIDES

Quality you can verify, documentation you can trust.

Browse our range of research compounds at veloxpeps.com. Every batch is HPLC-verified to $\geq 99\%$ purity and ships with a third-party Certificate of Analysis from Janoshik Analytical.

- **$\geq 99\%$ HPLC purity** — independently tested, CoA on every batch.
- **Fast UK dispatch** — orders out within 24 hours via Royal Mail Tracked 24.
- **Free UK shipping over £80** — with Pay by Bank at checkout.
- **Free tools** — including the reconstitution calculator at veloxpeps.com/tools/reconstitution-calculator/

For qualified researchers aged 21+

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